AI Evolution Capstone Project - Will coffee still be hot enough to drink when I get home from coffee shop?

Name	Sharon Maye
Student number	20122716
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Table of Contents

Declaration and AI-use statement ***2***

Summary - TODO ***2***

1. The Person and the Problem ***2***

2. The Law ***4***

Law in words **4**

Equation **4**

Law's Shape **4**

What the law leaves out **4**

Values **4**

Table 1: The constants in my law 4

3. The Answer Sheet ***5***

Exact Formula **5**

Solver **5**

Figure 1: The Answer Sheet **5**

4. The Model - TODO ***6***

5. The Comparison - TODO ***6***

6. Checks and Limits - TODO ***6***

7. Large Language Models and this Problem - TODO *7*

8. How I worked – AI Tools in my Workflow - TODO *7*

9. Where this is heading - TODO *7*

10. Conclusion - TODO *8*

References – In Progress *8*

Appendix A – In progress *8*

Notebooks within Repo **8**

Appendix B – In Progress *9*

Table 4 – AI Use Log **9**

Declaration and AI-use statement

This report and the work in my repository are my own. I used AI tools as described in section 8 and Appendix B, and I can explain every line I submit.

Summary - TODO

1. The Person and the Problem

From sample report

- *Who would use your answer, and how they work today.*
- *What your model would change for them.*
- *Your question in one sentence, and your success test as a number.*

Class example: A home baker deciding when the starter is ready to bake with; success: the height at 8 hours within 0.1 cm of the exact formula.

This project aims to determine if a coffee purchased in a takeaway cup would still be hot enough for a commuter to enjoy when they get home. It will help them determine if

the coffee will still be at an enjoyable temperature when they get home or if they should drink it before leaving the coffee shop as it will be too cold then they finish their commute.

The question it aims to answer is - Can a model that learns only from Newton's law of cooling, with no measured data, predict the time to reach your target temperature for a commuter deciding whether a takeaway coffee will still be enjoyable at home?

Success would be actual temperature when they get home being within 1 degree Celsius of the predicted temperature.

2. The Law

Law in words

Newton's Law of Cooling says that the hotter an object (coffee in our case) is compared with its surroundings , the faster it cools

Equation

How fast the temperature falls = k × (T – Ta)

Law's Shape

One thing changes over time

What the law leaves out

The law does not consider the following

- Type of coffee cup
- If there is a lid or not
- What type of coffee eg more milk may adjust temperature
- Changing ambient temperature eg car temperature may rise throughout the journey
- Coffee may cool quicker if aircon is on and blowing towards the coffee cup

Values

Starting Coffee Temperature: 55 Degrees Celcius

Range: We will look at the temperature drop from 0 to 60 minutes

Table 1: The constants in my law

What	Value	Unit	Source
Cooling Rate (k)	0.03	Per minute	U. H. Kurzweg (2017),"Cooling of a Cup of Coffee", University of Florida, p. 3 https://web.mae.ufl.edu/uhk/COOLING-COFFEE.pdf
Starting Coffee Temperature (T)	55	Degree Celcius	According to https://mtpak.coffee/2022/08/takeaway-cups-coffee-temperature-ideal-serving/ The ideal temperature for coffee in takeaway cups is 49 to 60 degrees

			Celsius so we will use 55 for testing purposes Can be requested by commuter to barista when the coffee is served
Ambient room temperature (Ta)	20	Degree Celcius	Average internal temperature of my person car while driving

3. The Answer Sheet

This was produced by running notebooks/capstone_step1.ipynb in Google Colab, see more details in Appendix A

Exact Formula

This is based on OpenStax, Calculus Volume 2, Section 2.8, "Exponential Growth and Decay", Example 2.45- https://openstax.org/books/calculus-volume-2/pages/2-8-exponential-growth-and-decay

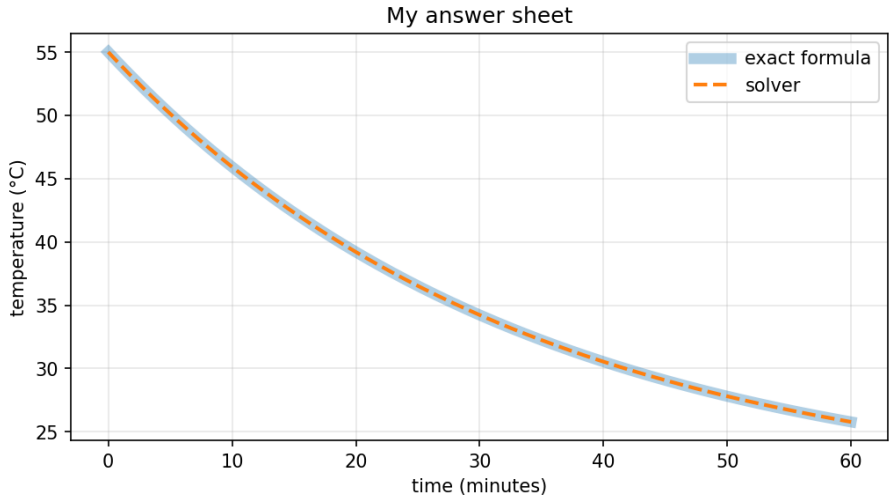
```
Ta + (T - Ta) * np.exp(-k * t)
```

Solver

This is based on the formula provided by lecturer in initial project feedback

```
-k * (y - Ta)
```

Figure 1: The Answer Sheet



4. The Model - TODO

From sample report

How it learns from the law: guess, check against the law at the checking points, blame, nudge; no data.

▪ *How the start is built in; the size of the network; the training settings.*

▪ *Figure: the loss curve. Figure: the model against the answer sheet.*

▪ *The error, and whether it passes your success test. Any problem you hit and how you fixed it.*

Figures & Tables – Figure 2 – The Loss curve, Figure 3 – Your model against the answer sheet

5. The Comparison - TODO

From Sample report

How the example-learner was set up: points from your answer sheet with a little noise; same size and training time.

▪ *Table: accuracy, time per answer and size for your model and each example-learner.*

▪ *Figure: error against the number of examples.*

▪ *The outside-the-range test, and the smallest version of your model that still passes your success test.*

▪ *Your answer to your comparison question, in one sentence with a number.*

Figures & Tables – Table 2 – The comparison, accuracy, time per answer and size, Figure 4 – Error against the number of examples

6. Checks and Limits - TODO

From sample report

Table: the three checks (new points; something that must always be true; the 20% wrong-law test) with

their results.

▪ *What each result means for your person.*

▪ *What the law leaves out, and what that means for someone relying on your model.*

Class example: Must always be true: the starter never rises above the top of the jar and never shrinks.

Figures & tables – Table 3 – The three checks and their results

7. Large Language Models and this Problem - TODO

From sample report

How a large language model works, in your own words.

▪ *What it would take to do your job with one: data, size, cost, accuracy; consider smaller and faster models*

(Weeks 8 and 9).

▪ *When an LLM would be the better choice, and when your model would.*

8. How I worked – AI Tools in my Workflow - TODO

From sample report

Which AI tools you used, and for what: understanding the law, writing and fixing code, tests, the README.

▪ *One example where a tool helped, and one where it was wrong, and how you found out.*

▪ *How these tools changed the way you worked. This must match your AI-use log.*

9. Where this is heading - TODO

From sample report

Is science-inspired AI real progress or hype for your person? Use Week 1's seven questions.

▪ *Assess at least one of edge computing, federated learning or quantum approaches for your use.*

10. Conclusion - TODO

From sample report

- *Your answer to your question, the main limit, and one thing you would do next.*

References – In Progress

Number/Claim	Reference
Cooling Rate per minute	U. H. Kurzweg (2017),"Cooling of a Cup of Coffee", University of Florida
Starting Coffee Temperature	https://mtpak.coffee/2022/08/takeaway-cups-coffee-temperature-ideal-serving/
Exact Formula	Strang, G., & Herman, E. (2016). Calculus Volume 2. OpenStax. Section 2.8: Exponential Growth and Decay. https://openstax.org/books/calculus-volume-2/pages/2-8-exponential-growth-and-decay

Appendix A – In progress

Repository - https://github.com/sharonjmaye/ai-evolution-2026-capstone

Final Commit – TBC

Notebooks within Repo

Notebook within Repo	Purpose, how to run and other important information eg random seeds
notebooks/capstone_step1.ipynb	Purpose - Generate answer sheet using solver and exact formula Run – Using Google Colab

Appendix B – In Progress

Table 4 – AI Use Log

Add details based on /Users/sharonmaye/SETU/Evolution/ai-evolution-2026-capstone/AI_USE_LOG.md

Date	Tool	What I asked	What I used And What I changed/checked
2026-11-07	ChatGPT	Help me find exact formula for Newon's Law of Cooling with Citatation	Used - The formula provided based on the citation | Checked the citation and that the formula was correct using Changed/Checked - Checked the citation and that the formula was correct in notebooks/capstone_step1.ipynb
2026-11-07	ChatGPT	Help me find cooling constant for rate of coffee cooling per minute with Citation	Used – The suggested cooling constant Changed/Checked - Checked the citation